What Does Feedstock Cost When Three Recycling Technologies Compete for the Same Waste?

A game-theory extension of the multilayer-film decision-support paper
(linear-cost model, base case and feedstock-scarcity sweep)

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The short version

The question. In the multilayer-film paper, one planner decides where all 8,000 t/yr of waste goes. Here we ask what happens if STRAP, incineration and gasification are three *separate firms* that must compete to buy that same limited waste. Competition puts an implicit price on the waste, called λ (USD per ton). It is not assumed; it comes out of the model.

What we found (base case, 8,000 t/yr):

- $\lambda = 175.57$ **USD/ton**, and incineration is the firm that sets it.
- STRAP wins the waste first: it takes its full 6,000 t/yr because it can afford to pay about 6.5 times more per ton than incineration. Incineration gets the remaining 2,000 t/yr. Gasification gets nothing.
- Gasification loses money on every ton it processes, even if the waste is free (net margin −16.69 USD/t), so it never enters at any waste volume.
- As the available waste changes, the price does not move smoothly. It takes only three values: 1,137.84, 175.57 or 0 USD/t, switching at 6,000 and 10,000 t/yr.

What we need from you. Several inputs are not in the paper and had to be assumed. The headline price is controlled almost entirely by *incineration's* numbers, so the three questions that matter most are **Q1** (incineration operating cost), **Q2** (incineration revenue) and **Q3** (plant capacities) in Section 6.

Color key. **[PAPER]** value taken from the paper; **[DERIVED]** value recovered from the authors' own notebook/spreadsheet (real data, not printed in the paper); **[ASSUMED]** value with no source, a defensible default that needs your check.

1 The idea in plain words

Three firms, one pile of waste. Each technology is treated as its own profit-maximizing firm: **S** (STRAP), **W** (incineration) and **G** (gasification for energy recovery). They all draw on the same 8,000 t/yr of post-industrial multilayer film. Every ton one firm takes is a ton the others cannot

have, and that shared limit is what couples them. Landfill is not a firm. It is only where each firm's leftover residue goes (at a fee) and where any unbought waste ends up.

The price that clears the market. Think of an auction for a fixed pile of waste. Each firm has a *walk-away price*: the most it could pay for one more ton and still break even on operations. We call this its **net margin** m_i. The market-clearing price λ is the price at which the firms' total demand just matches the pile. Economically, λ is the price of scarcity: what the waste is worth once firms have to compete for it.

Why the outcome is a queue. In this version every firm's operating cost is a fixed amount per ton. A firm that earns the same margin on every ton has no reason to take "a little": if the price is below its walk-away price it wants as much as it can process, and if the price is above, it wants none. So the firms line up in order of walk-away price, each takes its full capacity, and the last firm in line, the one only partly filled, sets the price. This is called a *merit order*, and it is why the answer can be found by simple arithmetic rather than a numerical solver.

2 What each firm earns per ton

Each firm's net margin is what is left of a ton's revenue after its operating cost and the landfill fee on its own residue (the feedstock itself is free in the base case, $p_{\text{raw}} = 0$):

$$m_i \;=\; \text{revenue}_i \;-\; \text{operating cost}_i \;-\; \text{landfill fee} \times \text{residue fraction}_i.$$

Table 1: Net margin per ton of feedstock, USD/ton. Landfill fee is 60 USD/t times the residue fraction.

Firm	Revenue	− Operating cost	− Landfill on residue	= Net margin m_i
S STRAP	1,468.39	311.28	19.26 (0.321 t/t)	**1,137.84**
W Incineration	259.57	75.00	9.00 (0.150 t/t)	**175.57**
G Gasification	147.98	159.87	4.80 (0.080 t/t)	**−16.69**

Three things stand out. STRAP's margin is large because resin is worth much more than energy. Incineration earns a thin but positive margin. And gasification's margin is negative: with the authors' own revenue figure (148 USD/t) and operating cost (160 USD/t), it costs more to run than the energy it produces is worth, before any feedstock price is even charged.

3 Result at the base case (8,000 t/yr)

Fill the queue in order of margin. STRAP takes its full 6,000 t/yr (it would take more if it could). That leaves 2,000 t/yr, which is less than incineration's 4,000 t/yr capacity, so incineration is only partly filled and is exactly on the margin. Gasification's margin is negative, so it takes nothing. The price is therefore incineration's walk-away price:

$$\boxed{\lambda^* = m_W = 175.57\ USD/\text{ton}}$$

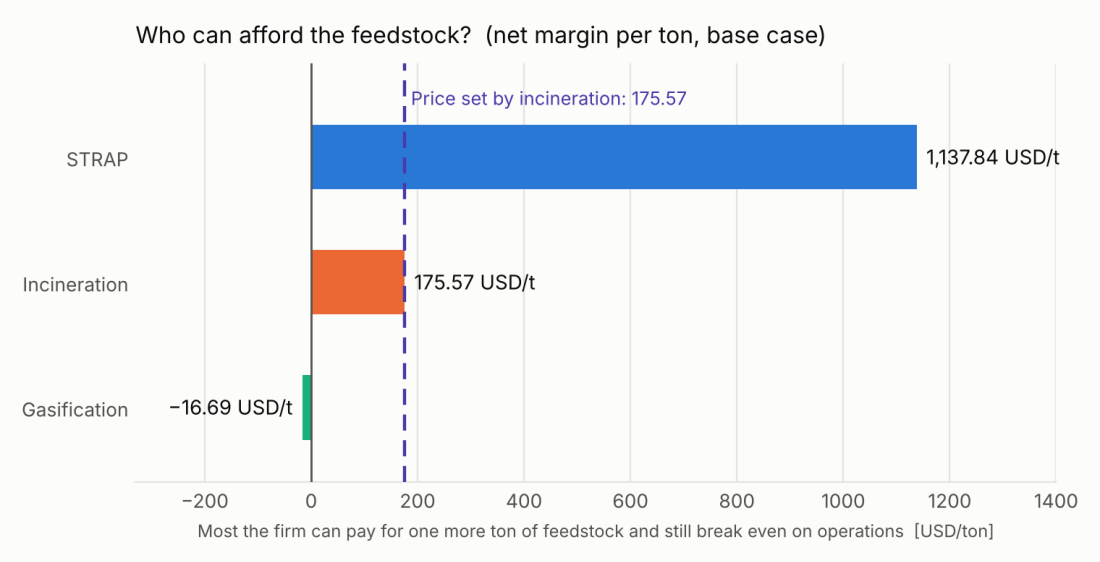


Figure 1: Each firm's walk-away price. The dashed line is the equilibrium price at 8,000 t/yr: STRAP is far above it, gasification is below zero, and incineration sits exactly on it.

Table 2: Base-case equilibrium. Operating profit is margin × tons; profit after capital subtracts the annualized capital charge (CRF = 0.1175).

Firm	Tons/yr	Share	Status	Operating profit (USD/yr)	Profit after capital (USD/yr)	Emissions (t CO_2e/yr)
S	6,000	75%	At capacity	6,827,062	523,742	7,329
W	2,000	25%	Sets the price	351,140	−3,173,860	4,786
G	0	0%	Priced out	0	−2,282,998	0
Total	8,000	100%		7,178,202	−4,933,116	12,115

How to read λ. Every firm pays $p_{\text{raw}} + \lambda$ per ton of waste, where p_{raw} is whatever the waste generator charges (zero here). Raising p_{raw} lowers λ one-for-one, so the total price a firm actually pays stays at 175.57 USD/t until p_{raw} reaches 175.57 and λ hits zero. STRAP's *capacity shadow price* is $\mu_S = m_S - \lambda = 962.27$ USD/(t/yr): how much STRAP's operating profit would rise for each extra ton/yr of plant, at an unchanged price.

Worth knowing: the price depends on incineration existing

Capital charges are treated as already paid, because capacities are fixed inputs. On that basis incineration runs, since it earns 351,140 USD/yr in cash. But after its *assumed* 30 million USD plant is paid for, it loses 3.2 million USD/yr, and only STRAP is profitable. If no incinerator were built at all, STRAP would still take 6,000 t/yr, 2,000 t/yr would go unbought, and λ at 8,000 t/yr would be **0**, not 175.57. The headline price therefore rests on the incinerator being there, which is one more reason Q1–Q3 matter.

4 What happens as feedstock availability changes

We re-solved the equilibrium for every availability Q_{tot} from 0 to 18,000 t/yr (207 points, including each switching point and points just on either side).

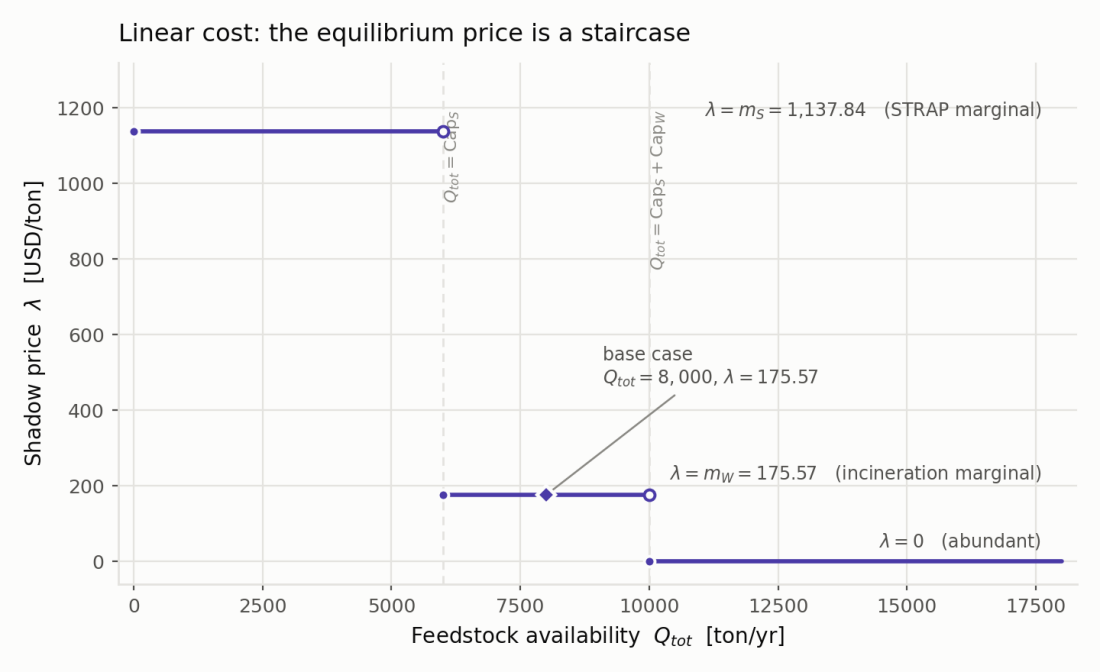


Figure 2: Equilibrium price against feedstock availability. It is a staircase: flat within a regime, with a jump each time the last firm in line changes.

Table 3: The three regimes.

Regime	Availability (t/yr)	λ (USD/t)	Who sets the price
Severe scarcity	below 6,000	1,137.84	STRAP
Moderate scarcity	6,000 to 10,000	175.57	Incineration
Abundance	above 10,000	0	Nobody; extra waste is landfilled

The switching points are exact: 6,000 is STRAP's capacity, and 10,000 is STRAP's plus incineration's. The upper limit is 10,000, not the 14,000 t/yr of total installed capacity, because gasification never runs.

Exactly at 6,000 and 10,000 the price is not unique. At $Q_{\text{tot}} = 6{,}000$ any price from 175.57 to 1,137.84 is consistent with equilibrium, and at 10,000 any price from 0 to 175.57. The tables and figures report the *lowest* consistent price at each of these two points. So the drop at 6,000 should be read as "just below versus just above," not as a response to a single extra ton. Just above 6,000 the price falls by 962.27 USD/t, an 85% drop, because the next buyer in line (incineration) values a ton at roughly one-sixth of what STRAP does.

What this can and cannot tell us. The order of the firms and the two switching points are robust: they depend only on how the margins rank and on the capacities. What the model cannot show is a *gradient*. At 6,100 t/yr and at 9,900 t/yr it returns the same price, 175.57. With a

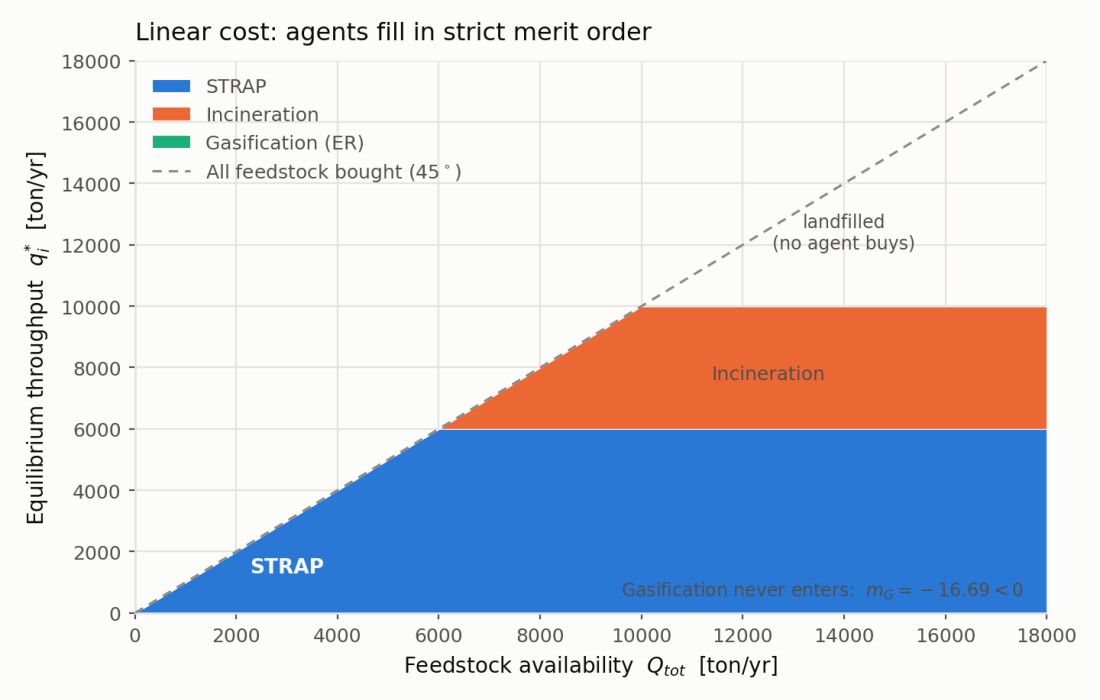


Figure 3: Who gets the waste. STRAP fills first, then incineration; gasification never enters. Above 10,000 t/yr the gap to the dashed line is waste nobody buys.

constant cost per ton, the model can describe scarcity regimes but not "a little scarcer." A convex (rising marginal cost) version would give a sloped price and is the natural next step.

5 What moves the answer, and what does not

At 8,000 t/yr the price is simply incineration's margin:

$$\lambda^* \;=\; m_W \;=\; \mathrm{REV}_W - a_W - p_{\mathrm{raw}} - c_{\mathrm{lf}}\,\rho_W \;=\; 259.57 - 75.00 - 0 - 60 \times 0.15 \;=\; 175.57.$$

That one line explains most of the sensitivity:

- **Moves it one-for-one:** incineration's revenue and operating cost, and the feedstock price p_{raw}.
- **Moves it through the landfill fee:** each extra 0.01 of incineration's residue fraction lowers λ by 0.60 USD/t, and each extra 10 USD/t of landfill fee lowers it by 1.50 USD/t.
- **Does not move it at all:** every STRAP economic number (as long as STRAP's margin stays above incineration's, i.e. STRAP revenue above about 506 USD/t), every gasification number (unless gasification revenue rose above about 340 USD/t, more than double today's), and the capital-recovery convention.
- **Moves it in jumps:** the plant capacities. They decide which regime 8,000 t/yr falls in (Table 3).

6 Questions for Paola

These are inputs that are not in the paper, ordered by how much they change the headline price. The last column shows what λ becomes at 8,000 t/yr if the value were different (everything else held at the base case).

	What we need from you	What we assumed	Effect on λ at 8,000 t/yr
Move the headline price directly			
1	**Incineration operating cost.** The workbook records it as zero (incineration is modeled only as an off-take market). Do you have a per-ton figure for this feedstock?	75 USD/t, mid-range US waste-to-energy O&M (typical 60–90) **[ASSUMED]**	60 gives 190.57; 90 gives 160.57. One-for-one.
2	**Incineration revenue.** Is `Cwte` = 259.57 USD/t the value of the energy sold per ton burned, and `Cgas_pw` = 110 USD/t the transfer price for the waste itself? We used 259.57 because an operating firm earns the product value.	259.57 USD/t **[DERIVED]**	If 110 were the right revenue, λ falls to **26.0** (about 6.8 times lower).
3	**Plant capacities.** Is there a defensible single-plant scale for STRAP, and any basis for incineration and gasification capacities?	6,000 / 4,000 / 4,000 t/yr **[ASSUMED]**	175.57 while $\text{Cap}_S < 8{,}000 < \text{Cap}_S + \text{Cap}_W$. 1,137.84 if $\text{Cap}_S > 8{,}000$; 0 if the two together are below 8,000.
4	**Landfill tipping fee.** The spreadsheet implies 7.836 USD/t, about 8 times below real Wisconsin gate rates. Is it a tipping fee or a different cost (e.g. processing)?	60 USD/t **[ASSUMED]** (paper: 7.836)	7.836 gives 183.39.
5	**Feedstock price.** What should a recycler pay for post-industrial waste? The paper gives it free; the authors' model implies 110 USD/t in another context; it could even be negative.	0 USD/t **[ASSUMED]**	110 gives 65.57; −50 gives 225.57. One-for-one, down.
6	**Ash/residue fractions.** The workbook records zero "waste to disposal" for incineration and gasification. Placeholder, or accounted for elsewhere?	0.15 and 0.08 t/t **[ASSUMED]**	With zero ash in incineration, 184.57; each extra 0.05 lowers λ by 3.00.
Matter only in other scenarios			
7	**Gasification revenue inputs.** The heat price (0.13 USD/MJ, i.e. 130 USD/GJ) and heat yield (583 MJ/t) both look unusual individually. Are they in the units the notebook implies?	147.98 USD/t **[DERIVED]**	None at 8,000 t/yr unless revenue exceeds about 340. It decides whether gasification ever runs: it needs above 164.67 (+11%) to break even on operations.

	What we need from you	What we assumed	Effect on λ at 8,000 t/yr
Do not change λ			
8	**Capital recovery convention.** We used 10%/20-year discounting (CRF 0.1175); the paper divides by 20 (0.05). Deliberate?	0.1175 [ASSUMED]	None. Changes only profit after capital.
9	**Emissions units.** Table 8's "kg CO_2-eq/year" reproduces exactly if the data are really tonnes. Can you confirm the Table 8–10 columns are tonnes?	Tonnes	None. Affects emission totals only.
10	**Table 8 profit.** We could not rebuild the published 7,551,142 USD/yr from the workbook's per-wash CAPEX/OPEX under any single convention (shredder charged once vs. twice; wash-2 capital in or out). Which convention produced it?	—	None. Only a plausibility check.

7 Limits of this version, and what we checked

Deliberate simplifications. Operating cost is linear (constant per ton), which is what makes the price a staircase. Product prices, the landfill fee and the feedstock price are fixed: no firm's actions move a market price. There is no emissions cap (emissions are reported, never constrained), and capacities are fixed inputs rather than the result of a capacity-choice game. A convex-cost version is the natural next step: it would give a smoothly varying price and a unique split of waste among firms.

Two caveats on the base-case profits. (i) STRAP's capital (53.6 million USD) comes from the 8,000 t/yr simulation, but we give STRAP a 6,000 t/yr capacity. Scaling capital down with the usual six-tenths rule would cut its capital charge from 6.30 to 5.30 million USD/yr and raise STRAP's profit after capital from 0.52 to about 1.5 million USD/yr. This does not touch λ. (ii) The incinerator's 30 million USD capital is assumed, not sourced, and is what makes it unprofitable after capital (see the box in Section 3).

How we checked the numbers. Every derived number in this report (revenues, margins, capital charges, emission factors and the base-case results) was re-derived independently from the raw inputs, without using the model code, and agrees with the code's output. The 207-point sweep was also recomputed point by point by brute-force search over prices, and every point matches. All equilibrium conditions hold with a worst violation of exactly zero (the solution is closed-form, so there is no iteration error), the price never increases as availability grows, and no two firms are ever tied on the margin at a swept point.

A Parameters

Table 5: Inputs, by firm. Units are per ton of feed unless stated.

Parameter	Value	Source
Feedstock		
Base availability	8,000 t/yr	[PAPER]
Mass fractions PE / PET / N6 / EVOH	0.60 / 0.20 / 0.10 / 0.10	[PAPER]
S — STRAP (frozen to the paper's best pathway: wash 1 PE/PCE, wash 2 EVOH/GBL)		
Yield (PE and EVOH recovered)	0.97	[PAPER]
Resin prices, PE / EVOH	1,173 / 8,100 USD/t	[PAPER]
Revenue per ton of feed	1,468.39 USD/t	[DERIVED]
Residue fraction (PET + N6 + losses)	0.321	[DERIVED]
Operating cost	2,490,259 USD/yr at 8,000 t/yr, i.e. 311.28 USD/t	[DERIVED]
Installed capital (wash 1 + wash 2 + shredder)	53,645,274 USD	[DERIVED]
Process emissions	1.194 t CO_2e/t	[DERIVED]
Capacity	6,000 t/yr	[ASSUMED]
W — Incineration		
Revenue (energy value)	259.57 USD/t	[DERIVED]
Operating cost	75 USD/t	[ASSUMED]
Residue (ash) fraction	0.15	[ASSUMED]
Installed capital	30,000,000 USD	[ASSUMED]
Process emissions	2.38 t CO_2e/t	[DERIVED]
Capacity	4,000 t/yr	[ASSUMED]
G — Gasification for energy recovery		
Revenue: 583.33 MJ/t × 0.13 USD/MJ + 724.37 kWh/t × 0.0996 USD/kWh	147.98 USD/t	[DERIVED]
Operating cost	159.87 USD/t	[DERIVED]
Residue (ash/char) fraction	0.08	[ASSUMED]
Annualized capital ($15{,}749.6 \times \text{Cap}^{0.6}$; already annualized by the authors)	2,282,998 USD/yr	[DERIVED]
Process emissions	1.07 t CO_2e/t	[DERIVED]
Capacity	4,000 t/yr	[ASSUMED]
Shared		
Feedstock price p_{raw}	0 USD/t	[ASSUMED]
Landfill tipping fee c_{lf}	60 USD/t	[ASSUMED]
Capital recovery factor (10%, 20 years)	0.1175 per year	[ASSUMED]
Landfill emission factor (0.0816 plus a 9.2-mile haul)	0.0865 t CO_2e/t	[DERIVED]

B Where the harder numbers come from

STRAP capital and operating cost. The paper prints neither. They come from the authors' BioSTEAM results for the best pathway under energy scenario 1. Capital is 35,957,532 (wash 1) + 14,557,669 (wash 2) + 3,130,073 (shredder). Operating cost is 941,557 + 479,709 + 1,068,992 USD/yr. The shredder is counted once; the authors' workbook charges it to both washes. The process emission factor is built from the per-resin factors (1.1326 for PE, 2.1661 for EVOH, 0.5575 for the shredder, all t CO_2e per ton of resin), scaled to the 4,656 t/yr of PE and 776 t/yr of EVOH recovered at 8,000 t/yr.

Incineration. The paper does not model incineration as a facility. The planner *sells* waste to it at 110 USD/t (`Cgas_pw`), and the workbook records its capital and operating cost as zero. Because the game needs incineration to be an operating firm, we used `Cwte` = 259.57 USD/t for revenue and assumed the operating cost and capital (Q1, Q2).

Gasification. Revenue is built from the authors' own yields and prices. Two inputs look off on their own: the heat price (130 USD/GJ) is roughly ten times typical industrial heat prices, and the heat yield (583 MJ/t) is far below the heating value of plastic (30–40 GJ/t). We cannot say which way correcting them would move revenue, because the two errors, if they are errors, push in opposite directions. This is Q7. Capital is already annualized in the authors' spreadsheet, so the capital recovery factor is not applied again.

Emissions units. Table 8 of the paper labels the landfill-only emissions "652.80 kg CO_2-eq/year." That is exactly $0.0816 \times 8{,}000$, and the underlying column is in tonnes, so we read the label as an error and apply no unit correction (Q9). Two small consistency notes: the landfill factor used for residue includes a 9.2-mile haul (0.0865), while incineration's process factor excludes its 151-mile haul; including it would add about 1.3% to system emissions.

Comparison with the paper. Routing all 8,000 t/yr to STRAP alone gives an operating profit of 9,102,749 USD/yr here. The paper's Table 8 maximum-profit figure is 7,551,142 USD/yr, but that figure is after capital charges and includes a gasification pathway, so the two are *not* like-for-like. The agreement in order of magnitude is only a rough plausibility check, not a validation.

C The math, for reference

Each firm $i \in \{S, W, G\}$ chooses throughput $q_i \geq 0$ (ton/yr) to maximize

$$\pi_i \;=\; m_i\, q_i \;-\; \mathrm{ACC}_i, \qquad m_i \;=\; \sum_k P_k\, \eta_{ik} \;-\; a_i \;-\; p_{\mathrm{raw}} \;-\; c_{\mathrm{lf}}\, \rho_i,$$

where P_k are product prices, η_{ik} the yield of product k per ton of feed, a_i the operating cost per ton, ρ_i the residue fraction and ACC_i the annualized capital charge (a constant, so it does not affect q_i^*). The firms are linked by one shared constraint:

$$q_i \leq \mathrm{Cap}_i \;\; (\text{multiplier } \mu_i), \qquad \sum_j q_j \leq Q_{\mathrm{tot}} \;\; (\text{multiplier } \lambda).$$

Because the constraint is shared, this is a *generalized* Nash game. We solve for the *variational equilibrium*, in which every firm faces the same λ, the natural concept for a market with a single

clearing price. Stacking the firms' optimality (KKT) conditions gives

$$\begin{aligned} 0 \le q_i \;&\perp\; \lambda + \mu_i - m_i \ge 0, \\ 0 \le \mu_i \;&\perp\; \mathrm{Cap}_i - q_i \ge 0, \\ 0 \le \lambda \;&\perp\; Q_{\mathrm{tot}} - \textstyle\sum_j q_j \ge 0. \end{aligned}$$

The term $\lambda + \mu_i - m_i$ does not depend on q_i, which is the formal statement that cost is linear. Each firm's best response is therefore a step:

$$q_i(\lambda) = \begin{cases} \mathrm{Cap}_i & m_i > \lambda, \\ 0 & m_i < \lambda, \\ \text{anything in } [0, \mathrm{Cap}_i] & m_i = \lambda. \end{cases}$$

Aggregate demand $D(\lambda) = \sum_{i:\, m_i > \lambda} \mathrm{Cap}_i$ is a step function, and the equilibrium price is the lowest $\lambda \ge 0$ with $D(\lambda) \le Q_{\mathrm{tot}}$; the firm exactly at λ, if any, absorbs the remainder. Firms strictly above it run at capacity, and firms strictly below take nothing. If two firms ever tied on margin, the price would still be unique but the split between them would not. That does not happen in the base case or anywhere in the sweep. System emissions are $\sum_i e_i q_i^* + g_{\mathrm{lf}} \left(Q_{\mathrm{tot}} - \sum_i q_i^*\right)$, reported only and never constrained.

Source paper. Muñoz-Briones, Ikegwu, Martinez, Granger, Van Lehn, Zavala, Huber and Avraamidou, *Decision support for circular economy waste management of mixed plastic streams: A case study on post-industrial multilayer films.*

Code. `Code/linear_model.py` (solver and sweep) and `Code/plots_linear.py` and `Code/plot_margins.py` (figures); full sweep output in `Code/results/linear_sweep_q_tot.csv`.